Table 1: Examples eliminated in preprocessing

Dataset	ϵ	Eliminated because robust to *all* models	Eliminated because NOT robust to *all* models
MNIST	0.1	97.67%	0.07%
	0.2	91.85%	0.42%
	0.3	69.69%	2.84%
CIFAR	2/255	65.60%	3.97%
	8/255	56.23%	14.11%

this is a provable upper bound of PGD attack error and has been used as the standard way to evaluate certified defense methods (Wong and Kolter, 2018; Wong et al., 2018). Since there is no existing work on boosting provable robustness, our baseline is the naive ensemble, where each model is equally weighted. Because our purpose is to show how optimally RobBoost weights each base model, we do not focus on tuning each base model to achieve state-of-the-art results on each dataset. We use small base models, where all MNIST models sum to 9.1 MB and all CIFAR models sum to 10.0 MB. We precompute $\widehat{\mathbf{L}}$ for all training examples for each model. This precomputation takes similar time as 1 epoch of robust training (Wong and Kolter, 2018), since they need to compute the same bounds every epoch for training a single robust model, and they typically need 100 to 200 epochs for training. The time of our experiments is dominated by training each base models (hours to days each) rather than precomputing these matrices and solving the ensemble objective (1-2 hours).

Data Elimination. We first remove all data points that have positive margins on *all* base models, as they will remain robust regardless of any positive weights. Similarly, we remove all data points that have negative margins on *all* base models, as the ensemble is not capable to improve robustness for them. These preprocessing steps allow us to focus on the data points whose robustness can be potentially enhanced, and also reduce the effective training data size. In Table 1, we report the percentage of examples eliminated in this preprocessing step. For all models, a large portion of examples (ranging from 70% to 97%) can be excluded from the optimization step, greatly improving the efficiency of RobBoost.

Ensemble of robustly trained models with different architectures. We train 12 MNIST models and 11 CIFAR models with a variety of architectures. The models are trained using convex adversarial polytope (Wong et al., 2018), a certified defense method, under different ℓ_∞ norm perturbations ($\epsilon = \{0.1, 0.2, 0.3\}$ for MNIST and $\epsilon = \{\frac{2}{255}, \frac{8}{255}\}$ for CIFAR). Even the largest CIFAR model used is much smaller than the best ResNet model in Wong et al. (2018) (40 MB), as we desire to use an ensemble of small models to obtain better robustness than a single large model. For CIFAR, the smallest model is around 0.1 MB and the largest is 3 MB; see more details on model structure in Appendix B. We list results in Table 2. We can observe that RobBoost consistently outperforms naive averaging ensemble in both verified and clean error. Also the ensemble model we created performs better in all metrics than the single large model reported in literature (Wong et al., 2018).

In Figure 3, we plot the distributions of lower bounds of margins $M(\boldsymbol{x}_k, j)$ for two models. A value less than 0 indicates that an example cannot be certified. Compared to the naive ensemble, we can clearly observe that the distributions of margins for RobBoost ensembles have more mass on the positive side, reflecting the improvements on certified robustness.

Ensemble of the same model architecture with feature subsampled data. A common practice in building traditional ensemble models like random forest is to use feature subsampling, i.e., each base model only uses a subset of the features to train. In this experiment, we use only 1 model structure but randomly sample 80% pixels to train the model (a recent work (Hosseini et al., 2019) presented a similar idea, but their method is not a certified defense). We train 5 feature sub-sampled models for MNIST with $\epsilon = 0.3$ and CIFAR with $\epsilon = \frac{2}{255}$. In Table 3, we list the verified error and clean error of each model for both the naive ensemble and RobBoost ensemble. Since there are only 5 base models, RobBoost ensembles provide a small but consistent performance advantage in verified error.

Gradient Boosting of Robust Ensemble. Unlike previous experiments where all base models are given and fixed, we follow Eq. (23) and train models incrementally. We use cross-entropy loss and train an ensemble of 5 models on MNIST with $\epsilon = 0.3$ and CIFAR with $\epsilon = \frac{2}{255}$ in Figure 2. We observe that usually the first 2 or 3 models decrease verified errors most (on MNIST from 39.8% to 34.6%, and on CIFAR from 51.18% to 49.55%. It is challenging to further decrease this error with more models, but gradient boosting allows us to fix the data points lacking of robustness on previous models, achieving better performance than Table 2.

5 Conclusion

We propose the the first ensemble algorithm, RobBoost, to enhance *provable* model robustness by optimally weighting each base model. Our algorithm involves optimizing a surrogate of the lower bound of classification margin through the proposed coordinate descent algorithm, and consistently outperforms a naive averaging ensemble as well as a state-of-the-art single model certified defense in verified error and clean accuracy.

Table 2: Comparison of verified error, clean error and PGD error of RobBoost and naive ensembles. As a comparison to single model defense, for MNIST $\epsilon = 0.3$, the best model in Wong et al. (2018) achieves 43.10% verified error and 14.87% clean error; for CIFAR $\epsilon = \frac{8}{255}$, the ResNet (best) model in Wong et al. (2018) achieves 78.22% verified error and 71.33% clean error. All our CIFAR models combined have a smaller size (10 MB) than the ResNet model in Wong et al. (2018) (**40 MB**), and the RobBoost ensemble outperforms the single largest model in Wong et al. (2018) in both clean and verified error.

Model	ϵ	Ensemble	Verified Error		Clean Error	PGD Error	Ensemble
			Train	Test	Test	Test	Model Size
MNIST	0.1	Naive	3.55%	4.56%	0.39%	1.56%	9.1M
		RobBoost	**2.65%**	**4.47%**	0.39%	**1.17%**	**5.3M**
	0.2	Naive	13.82%	13.66%	2.34%	3.91%	9.1M
		RobBoost	**12.41%**	**12.58%**	**1.95%**	**3.12%**	**7.4M**
	0.3	Naive	39.22%	38.60%	11.33%	19.92%	9.1M
		RobBoost	**37.42%**	**36.61%**	**9.77%**	**19.53%**	**5.7M**
CIFAR	2/255	Naive	50.02%	51.68%	38.28%	45.31%	10.0M
		RobBoost	**47.27%**	**49.51%**	**34.77%**	**41.80%**	**6.1M**
	8/255	Naive	74.95%	74.50%	62.89%	71.09%	10.0M
		RobBoost	**74.56%**	**74.27%**	**60.94%**	**69.14%**	**5.6M**

Table 3: Naive and RobBoost ensembles on 5 small models with the same architecture but trained on different feature subsamples. Since they have the same architecture, their robustness are similar and the optimal weight for each model is close to $\frac{1}{5}$; in this case the naive baseline is close to RobBoost as it is a special case of RobBoost.

Dataset	ϵ	Test Error	Base Models No.					Naive	RobBoost
			1	2	3	4	5		
MNIST	0.3	Clean	12.96%	13.99%	23.88%	12.77%	18.50%	13.34%	**12.51%**
		Verified	42.98%	45.3%	51.62%	45.13%	49.32%	43.94%	**42.84%**
CIFAR	2/255	Clean	42.33%	42.15%	41.77%	42.43%	41.86%	40.79%	**40.77%**
		Verified	55.00%	55.11%	54.97%	54.73%	55.07%	53.61%	53.61%

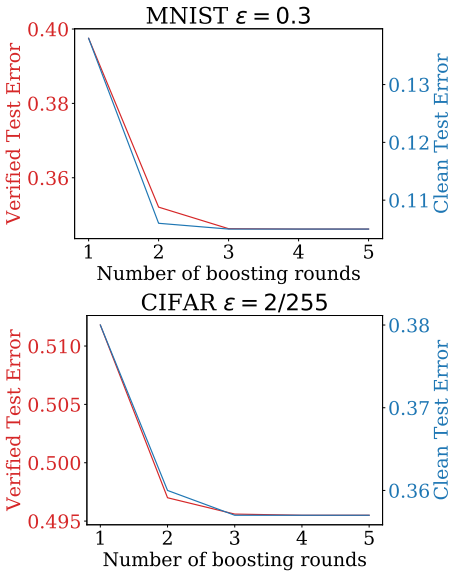

Figure 2: RobBoost in gradient boosting setting. With 3-5 base models RobBoost significantly reduces both verified and clean errors on both CIFAR and MNIST datasets compared to a single base model.

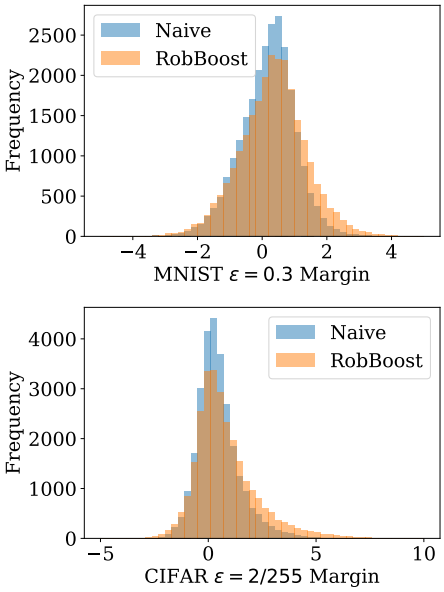

Figure 3: Distribution of margins' lower bounds. RobBoost moves the margins towards the positive side; a margin greater than 0 indicates that an example is certifiably robust.

References

Abbasi, M. and Gagné, C. (2017). Robustness to adversarial examples through an ensemble of specialists. *arXiv preprint arXiv:1702.06856*.

Athalye, A., Carlini, N., and Wagner, D. (2018). Obfuscated gradients give a false sense of security: Circumventing defenses to adversarial examples. *International Conference on Machine Learning (ICML)*.

Carlini, N. and Wagner, D. (2017a). Adversarial examples are not easily detected: Bypassing ten detection methods. In *Proceedings of the 10th ACM Workshop on Artificial Intelligence and Security*, pages 3–14. ACM.

Carlini, N. and Wagner, D. (2017b). Magnet and" efficient defenses against adversarial attacks" are not robust to adversarial examples. *arXiv preprint arXiv:1711.08478*.

Carlini, N. and Wagner, D. (2017c). Towards evaluating the robustness of neural networks. In *2017 38th IEEE Symposium on Security and Privacy (SP)*, pages 39–57. IEEE.

Dvijotham, K., Garnelo, M., Fawzi, A., and Kohli, P. (2018a). Verification of deep probabilistic models. *CoRR*, abs/1812.02795.

Dvijotham, K., Gowal, S., Stanforth, R., Arandjelovic, R., O'Donoghue, B., Uesato, J., and Kohli, P. (2018b). Training verified learners with learned verifiers. *arXiv preprint arXiv:1805.10265*.

Dvijotham, K., Stanforth, R., Gowal, S., Mann, T., and Kohli, P. (2018c). A dual approach to scalable verification of deep networks. *UAI*.

Dvijotham, K. D., Stanforth, R., Gowal, S., Qin, C., De, S., and Kohli, P. (2019). Efficient neural network verification with exactness characterization. *UAI*.

Ehlers, R. (2017). Formal verification of piece-wise linear feed-forward neural networks. In *International Symposium on Automated Technology for Verification and Analysis*, pages 269–286. Springer.

Freund, Y. and Schapire, R. E. (1997). A decision-theoretic generalization of on-line learning and an application to boosting. *Journal of computer and system sciences*, 55(1):119–139.

Friedman, J., Hastie, T., Tibshirani, R., et al. (2000). Additive logistic regression: a statistical view of boosting (with discussion and a rejoinder by the authors). *The annals of statistics*, 28(2):337–407.

Friedman, J. H. (2001). Greedy function approximation: a gradient boosting machine. *Annals of statistics*, pages 1189–1232.

Friedman, J. H. (2002). Stochastic gradient boosting. *Computational Statistics & Data Analysis*, 38(4):367–378.

Gehr, T., Mirman, M., Drachsler-Cohen, D., Tsankov, P., Chaudhuri, S., and Vechev, M. (2018). AI 2: Safety and robustness certification of neural networks with abstract interpretation. In *2018 IEEE Symposium on Security and Privacy (SP)*.

He, W., Wei, J., Chen, X., Carlini, N., and Song, D. (2017). Adversarial example defenses: ensembles of weak defenses are not strong. In *Proceedings of the 11th USENIX Conference on Offensive Technologies*, pages 15–15. USENIX Association.

Hein, M. and Andriushchenko, M. (2017). Formal guarantees on the robustness of a classifier against adversarial manipulation. In *Advances in Neural Information Processing Systems (NIPS)*, pages 2266–2276.

Hosseini, H., Kannan, S., and Poovendran, R. (2019). Dropping pixels for adversarial robustness. *arXiv preprint arXiv:1905.00180*.

Kariyappa, S. and Qureshi, M. K. (2019). Improving adversarial robustness of ensembles with diversity training. *arXiv preprint arXiv:1901.09981*.

Katz, G., Barrett, C., Dill, D. L., Julian, K., and Kochenderfer, M. J. (2017). Reluplex: An efficient smt solver for verifying deep neural networks. In *International Conference on Computer Aided Verification*, pages 97–117. Springer.

Liu, C., Arnon, T., Lazarus, C., Barrett, C., and Kochenderfer, M. J. (2019). Algorithms for verifying deep neural networks. *arXiv preprint arXiv:1903.06758*.

Liu, X., Cheng, M., Zhang, H., and Hsieh, C.-J. (2018). Towards robust neural networks via random self-ensemble. In *European Conference on Computer Vision*, pages 381–397. Springer.

Madry, A., Makelov, A., Schmidt, L., Tsipras, D., and Vladu, A. (2018). Towards deep learning models resistant to adversarial attacks. In *International Conference on Learning Representations*.

Mirman, M., Gehr, T., and Vechev, M. (2018). Differentiable abstract interpretation for provably robust neural networks. In *International Conference on Machine Learning*, pages 3575–3583.

Pang, T., Xu, K., Du, C., Chen, N., and Zhu, J. (2019). Improving adversarial robustness via promoting ensemble diversity. *arXiv preprint arXiv:1901.08846*.

Qin, C., Dvijotham, K. D., O'Donoghue, B., Bunel, R., Stanforth, R., Gowal, S., Uesato, J., Swirszcz, G., and Kohli, P. (2019). Verification of non-linear specifications for neural networks. *ICLR*.

Raghunathan, A., Steinhardt, J., and Liang, P. (2018a). Certified defenses against adversarial examples. *In-*